

Process Characterization Plan

**A-Mab Drug Substance — Cation Exchange Chromatography (Step 7) —
PCP-007**

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

ALCOA+, attributable, legible, contemporaneous, original and accurate, plus complete, consistent, enduring and available; AMV, analytical method validation; CEX, cation

exchange chromatography; Cpk, one-sided process capability index; CQA, critical quality attribute; DNA, deoxyribonucleic acid; DoE, design of experiments; DS, drug substance; ELISA, enzyme-linked immunosorbent assay; HCP, host cell protein; HMW, high molecular weight; ICH, International Council for Harmonisation; LOQ, limit of quantitation; MSAT, manufacturing science and technology; NOR, normal operating range; OD, optical density; PAR, proven acceptable range; PRESS, predicted residual error sum of squares; qPCR, quantitative polymerase chain reaction; RPN, risk priority number; RSD, relative standard deviation; RSM, response surface methodology; SEC-HPLC, size exclusion high performance liquid chromatography; SOP, standard operating procedure.

1 Purpose and scope

This plan defines the Stage 1 characterization study for the cation exchange chromatography step of the A-Mab drug substance process. It states which parameters will be varied, over which ranges, on which model of the commercial column, and how the resulting data will be analysed. The acceptance and decision criteria are fixed here, before any data exist (Section 7). The executed study and its conclusions are reported in PCR-007.

The step is operated in bind and elute mode on a strong cation exchange resin. Its principal duty is the reduction of high molecular weight aggregate, and it also delivers a large part of the clearance of host cell protein (HCP), residual DNA and leached Protein A. The step forms no quality attribute of its own. Every attribute in scope is established upstream, and every acceptance criterion used in this plan is a drug substance criterion (Table 4.1) that applies at the end of the process.

The study covers the 5 process parameters of the step, of which 4 will be varied together in designed experiments and 1 will be assessed one at a time, over the ranges given in Section 6. All runs will be executed on a scale-down model qualified against commercial scale data under SOP-1001, using a representative neutralized pool from the low pH viral inactivation step as feed. The measured responses are the pool aggregate, the pool HCP and the step yield. The work supports the transfer of the drug substance process from the development site to the commercial facility described in PTP-001, and it follows the lifecycle approach to process validation (U.S. Food and Drug Administration 2011) and the enhanced development approach of ICH Q8 and ICH Q11 (International Council for Harmonisation 2009, 2012).

Three things are outside this plan.

- Resin life cycle, packing and sanitization. These are qualified and controlled separately under SOP-2008, and cycle number is held constant across the study.
- Viral clearance. Cation exchange contributes to the removal of enveloped virus, but no clearance claim is made for this step. The modular claim rests on the low pH hold (PCP-006), anion exchange (PCP-008) and virus filtration (PCP-009), and is consolidated in PCMR-001 (International Council for Harmonisation 2023a).
- Confirmation at commercial scale. The ranges established here are confirmed during Stage 2, in the process performance qualification batches, and not by this study (U.S. Food and Drug Administration 2011).

2 Objectives

The study has five objectives.

- (i) Quantify the relationship between each of the 5 process parameters and the pool aggregate, the pool HCP and the step yield, over the characterization ranges given in Section 6.
- (ii) Identify which parameters and which two factor interactions govern each response, and separate them from those that do not.
- (iii) Establish the multivariate operating region over which pool aggregate remains within its drug substance criterion, and the proven acceptable range of each parameter for each governed attribute.
- (iv) Provide the evidence needed to classify every parameter of the step under SOP-4001.
- (v) Justify the normal operating ranges to be applied at commercial scale, and define the contribution this step makes to the control strategy.

The first three objectives are met by the designed experiments described in Section 6 and by the analysis defined in Section 5.5. The last two are met from those results and are delivered in PCR-007, which rolls up into PCMR-001.

3 Related documents

This plan sits under the transfer plan, the pre-characterization risk assessment and the characterization master plan. Its parents, its paired report and the document it rolls up into are listed in Table 3.1.

Table 3.1: Related documents in the A-Mab characterization package.

Docu- ment ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-007	Process Characterization Report (Cation Exchange Chromatography (Step 7))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The controlled procedures and the validated analytical methods that govern the study are listed in Table 3.2. Method performance is held in the validation reports and is summarized here only where it bears on the design (Section 5.3).

Table 3.2: Controlled procedures and validated analytical methods for the study.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2010	Operation of the Cation-Exchange Polishing Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3011	Size-Variants (SEC-HPLC)	Method validation

Reference	Title	Type
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

The cation exchange step is the polishing operation of a well established platform purification train. The load is the neutralized pool of the low pH viral inactivation step (PCP-006). It is applied to a strong cation exchange resin at a conductivity low enough for the antibody to bind, and impurities that bind weakly pass into the flow through or are removed in the wash. The antibody is then recovered with a pH step elution, and the pool is collected on ultraviolet absorbance between a fixed start criterion on the ascending edge of the peak and a stop criterion on the descending edge. Charge variants and glycosylation variants are not discriminated by the step under platform conditions (CMC Biotech Working Group 2009).

The platform train has been used to manufacture three earlier humanized IgG1 antibodies (X-Mab, Y-Mab and Z-Mab) and to supply the A-Mab clinical programme. The resin, the buffer system and the elution mode are unchanged from those products, and the mechanism of separation is the same. That experience shows that the performance of the step is governed by a small number of parameters, and it showed no effect of the remaining parameters across the ranges those earlier campaigns covered. For this reason the study confirms and bounds platform behaviour, and it is not a search for the operating conditions of the step.

Four mechanisms are expected to matter, and they set the factor list of the multivariate design.

- Protein load determines how much of the capacity of the column is used. As the load rises, weakly bound impurities are displaced from the resin into the product peak, so both aggregate clearance and HCP clearance are expected to fall. Yield is expected to fall at the top of the range through breakthrough.
- Load and wash conductivity sets the strength of the electrostatic competition at the resin surface. A higher conductivity is expected to weaken the binding of the more weakly retained HCP species, so that more of them are removed in the flow through and the wash. The effect RA-001 records for this parameter is a loss of HCP clearance (Table 4.2), and on that mechanism the risk sits at the bottom of the range, where the competition is too weak to displace those species.
- Elution buffer pH sets the strength of the elution. Aggregate carries a higher charge density than the monomer and is retained more strongly, so a stronger elution is expected to carry more aggregate into the collected pool.
- The stop collect criterion sets how much of the descending edge of the elution peak enters the pool. Aggregate is enriched in the trailing edge of a cation exchange elution, so a criterion that admits more of that edge is expected to

raise the aggregate of the pool. RA-001 records the effect as a minor increase (Table 4.2). Whether the criterion also costs yield is one of the questions the design answers, because step yield is measured on every run.

Elution flow rate acts on the separation through residence time and pressure drop, and not through the chemistry that governs binding and elution. It is not expected to interact with the four factors above, and it is assessed one at a time (Section 6.4).

4.2 Quality attributes in scope

The step governs four attributes of the drug substance and sets none of them. Each is listed in Table 4.1 with its acceptance criterion, its criticality and its Tool #1 score. Criticality is treated as a continuum, and the attributes scored high or very high are the ones called critical (International Council for Harmonisation 2023b; CMC Biotech Working Group 2009).

Table 4.1: Quality attributes this step governs, with acceptance criteria and criticality.

CQA	Category	Acceptance	Criticality	Tool #1
Aggregates (HMW)	size variant	0-5 % HMW (SEC)	H	60
Host Cell Protein (HCP)	process impurity	0-100 ng/mg	M-H	36
Residual DNA	process impurity	0-0.001 ng/dose	VL	6
Leached Protein A	process impurity	0-5 ppm	L-M	16

Aggregate is the attribute this step governs most directly. Cation exchange is the last step of the train at which high molecular weight aggregate is reduced, so the pool this step delivers carries the aggregate content of the drug substance. The acceptance criterion is not more than 5 % HMW (SEC) and the criticality is high (Table 4.1). The low pH hold immediately upstream generates aggregate, so the material presented to this step carries the highest aggregate burden anywhere in the train, and the clearance the step achieves is what stands between that burden and the specification.

HCP is of moderate to high criticality and is cleared by three steps in series. Protein A capture removes the bulk of it (PCP-005), cation exchange removes a further large fraction, and anion exchange polishes what remains (PCP-008). The drug substance criterion of not more than 100 ng/mg therefore applies to the output of the whole train and not to the pool of one intermediate step. Section 7 states how that distinction is handled in the decision criteria, because it is the one place where this plan does not judge a response against the criterion in Table 4.1.

Residual DNA and leached Protein A are of low criticality. Both are cleared here by a mechanism that prior knowledge treats as a fixed log reduction, and neither is

expected to depend on the factors of the design. Both are measured on every elution pool so that the clearance of the step is documented, but neither is a response of the designed experiments.

The charge variant and glycan attributes are formed in the production bioreactor (PCP-003) and are not modified by charge based purification (CMC Biotech Working Group 2009). They are not responses of this study, and the assumption that the step leaves them unchanged is recorded in Section 12.

4.3 Risk-based prioritization of parameters

RA-001 assessed every parameter of the step before this plan was written. It records for each parameter a prospective failure mode, the effect that failure would have on the product or on the process, and an initial severity, occurrence and detection score whose product is the initial risk priority number. The study type then follows from what the parameter can affect. A parameter that can affect a quality attribute is carried into the designed experiment of the step, and a parameter that acts on process performance alone is assessed one at a time. The effects and the scores RA-001 assigned to this step are given in Table 4.2.

Table 4.2: Pre-characterization effects, risk scores and assigned study type for the parameters of the step (RA-001).

Parameter	Effect	Sev.	Occ.	Det.	Initial RPN	As-signed study
Protein load	Reduced aggregate and HCP clearance (interactions with pH and wash conductivity)	8	7	8	448	multi-variate DoE
Load/Wash conductivity	Reduced HCP clearance	6	7	8	336	multi-variate DoE
Elution buffer pH	Alters aggregate distribution and yield	8	7	8	448	multi-variate DoE
Elution stop collect	Minor increase in pool aggregate	8	7	8	448	multi-variate DoE
Elution flow rate	Affects peak shape and yield (minor aggregate)	4	4	6	96	univariate

The outcome for this step is 4 parameters carried into a multivariate design and 1 assessed one at a time, which is the split shown in the study type column of Table 6.1. The highest initial risk priority number of the step is 448, and 3 of the 5 parameters reach it, because each of them is linked to the aggregate attribute and inherits its

severity. The elution flow rate is the only parameter scored on process performance alone, at 96, and it is the one assessed univariately. No parameter of the step was excluded from study altogether.

The ranges to be studied are wider than the ranges the commercial process will operate in. The characterization range of the protein load is 2.1 times the width of its normal operating range, and the range of the load and wash conductivity is 2.0 times its own. Widening the studied range serves three purposes. It gives the fitted model enough signal to resolve an effect against assay noise, it supports later movement inside the design space, and it locates an edge of failure where one exists inside the knowledge space (CMC Biotech Working Group 2009).

The elution pH range is the narrowest of the four in relative terms, at 1.3 times the width of its normal operating range. Elution pH drives the elution itself, and the platform buffer system holds it inside a narrow band. A wider excursion would change the character of the separation instead of probing its edges, and the resulting model would describe a step that the commercial process would never run.

5 Materials and methods

5.1 Scale-down model and its qualification

Every run of this study will be executed on a scale-down model of the commercial column, and the model will be qualified before the first designed experiment starts. The model is built on the standard principles of chromatography scaling. It uses the same resin, the same bed height and the same linear velocity as the commercial column, load is expressed in grams of product per litre of resin, and the wash and elution volumes are normalized to column volumes. Buffers are prepared to the same specification and from the same recipes as at scale.

Qualification follows SOP-1001. Triplicate runs at the set-point of every parameter will be compared with commercial scale data for product yield, pool volume, aggregate clearance, HCP clearance and leached Protein A. Column efficiency will be measured as plate count and peak asymmetry and judged against the packing criteria of SOP-2008. The acceptance criteria for qualification are stated in Section 7, and the qualification outcome is reported in PCR-007 before any characterization result.

Qualification matters because the conclusions of this study are applied directly to the commercial process. A range proven on the model becomes a range the commercial step may operate in, and a design space derived from the model is defended at the commercial scale. The model is representative for the attributes it was qualified against, at the set-point, and the study assumes that the same representativeness holds across the characterization ranges. That assumption is the principal limitation of the work and is recorded in Section 12.

The centre point of each design sits at the set-point of every factor. Its replicates give the pure error of the whole measurement chain, from column operation through sampling to assay, and that term is the reference against which lack of fit is judged (Section 5.5).

5.2 Operation

The step will be operated as it is at commercial scale, following SOP-2010. A cycle consists of equilibration, load, wash, elution and post-use treatment. The column is equilibrated in the load buffer at the conductivity of the run, the neutralized viral inactivation pool is applied to the target protein load, and the column is washed with the same buffer. The antibody is recovered with a pH step elution at the elution pH of the run. Pool collection starts at a fixed absorbance on the ascending edge of the

peak and stops at the stop collect criterion of the run. The column is then stripped, sanitized and stored under SOP-2008.

Inputs held to platform specification are controlled by identity and quality testing and are not characterized. These are the buffer compositions, the resin lot and the packing quality of the column. Chromatography is performed in a temperature controlled chamber, and the chamber temperature is recorded for every run, so that any excursion is available to the analysis as a covariate.

The feed for every run is a neutralized low pH inactivation pool. Each feed lot will be assayed for protein concentration, aggregate and HCP before use, and one lot will be used for all runs of a design wherever the available material allows it. Feed lots are recorded per run, and the lots used are reported in PCR-007.

5.3 Analytical methods

Every response and every attribute in scope is measured with a validated method, and the methods are listed with their validation reports in Table 3.2. Aggregate is measured by size exclusion chromatography (AMV-3011) and reported as the percentage of high molecular weight species. HCP is measured by enzyme-linked immunosorbent assay (AMV-3012) and reported against the product concentration in nanograms per milligram. Residual DNA is measured by quantitative PCR (AMV-3014) and leached Protein A by ELISA (AMV-3016). Product concentration is measured by ultraviolet absorbance. Full validation data for each method are in its validation report and are not restated here.

Two of these methods quantify an impurity in the elution pool, and their validated performance is given in Table 5.1. Performance for the remaining methods is held in their own validation reports and is not reproduced here.

Table 5.1: Validated performance of the two impurity methods that quantify pool results.

Method	Title	Precision (%RSD)	LOQ	Accuracy (%)	Variance share
AMV-3012	Host-Cell Protein by ELISA	9.5	1 ng/mg	95	0.4
AMV-3016	Leached Protein A by ELISA	6.5	0.25 ppm	97.4	0.41

The HCP assay has an intermediate precision of 9.5 %RSD and a limit of quantitation of 1 ng/mg, and about 40 % of the variance of a pool HCP result is attributable to the method itself. Pool HCP is expected to be the least reproducible of the three responses. The designs carry replicated centre points so that this variance is estimated directly and is not mistaken for lack of fit, and near-limit differences in HCP will be interpreted against the precision of the assay and not as process effects.

The leached Protein A assay is in the table for a different reason. Leached Protein A is not a response of the designed experiments, but it is measured on every elution pool, so its precision of 6.5 %RSD and its limit of quantitation of 0.25 ppm bound what a pool result can show. The method accounts for 41 % of the variance of a leached Protein A result, so a single pool value is read as a check on the clearance of the step and not as the measure of an effect.

5.4 Sampling plan

Samples are taken at the same points in every run, so that every run yields the same set of results.

- The load is sampled once per run and assayed for protein concentration, aggregate and HCP. This fixes the input to the step for that run and allows the clearance to be expressed as a fold reduction.
- The flow through and the wash are collected and sampled as one composite per run and assayed for protein concentration, which supports the mass balance of the cycle.
- The elution pool is sampled once per run and assayed for protein concentration, aggregate, HCP, residual DNA and leached Protein A. The pool results for the first three of these are the responses of the designed experiments.

47 runs of the designed experiments will be sampled in this way, 19 in the screening design and 28 in the response surface design. Of these, 3 and 4 respectively are centre point replicates at the set-point of every factor. The univariate runs of Section 6.4 and the qualification runs of Section 5.1 are sampled at the same points and assayed by the same methods, and they are additional to that count. Each sample is registered against the run that produced it at the time of sampling. Pool samples from one design will be analysed together in a single analytical campaign wherever sample stability allows, so that any drift in the assay does not align with the order of the runs.

5.5 Statistical methods

The analysis is defined here and will not be changed after the data have been seen. It follows SOP-1002.

Factors are coded so that the set-point is 0 and the two edges of the characterization range are -1 and +1. Coding puts every estimated effect on the same scale, and it makes the coefficients of a fitted model comparable across factors that are measured in different units.

The screening data will be fitted by ordinary least squares with all 4 main effects and all 6 two factor interactions. An effect is reported as twice the coded coefficient, which is the change in the response between the low and the high setting of the factor. Significance is judged at $\alpha = 0.05$.

The response surface data will be fitted with the full quadratic model in the 4 factors, which carries 15 terms including the intercept. This model, and not the screening model, is the predictive model of the step. The design space and the proven acceptable ranges will both be derived from it. The screening design identifies which factors are active and in which direction they act. It sets each factor at two levels, so it holds no information between those levels and cannot estimate curvature. Its coefficients are not used for prediction.

Model adequacy will be judged from R^2 , adjusted R^2 , the predicted R^2 obtained from the PRESS statistic, the overall F test, and an analysis of variance that partitions the residual sum of squares into lack of fit and pure error. The pure error term comes from the replicated centre points, so lack of fit is judged against the reproducibility of the process and the assay together. Residuals will be inspected against the predicted values and on a normal probability plot, and a model whose residuals show structure will not be used to define an operating region.

Capability at the drug substance will be estimated by Monte Carlo simulation of 2,000 batches with every process parameter varying inside its normal operating range. It is reported for each attribute in scope as a one-sided Cpk together with the margin between the simulated distribution and the limit. Capability estimated this way is a property of the whole train and of the fitted models, so the contribution of this step is stated alongside the contributions of the steps that share the attribute, and never in place of them.

6 Study design

6.1 Factors, ranges and study type

The parameters of the step, their set-points, the ranges to be studied and the normal operating ranges are given in Table 6.1.

Table 6.1: Process parameters, characterization ranges, normal operating ranges and study type.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Protein load	g/L resin	25	10-40	18-32	multivariate
Load/Wash conductivity	mS/cm	5	3-7	4-6	multivariate
Elution buffer pH	pH	6	5.8-6.2	5.85-6.15	multivariate
Elution stop collect	OD desc.	1	0.5-1.5	0.7-1.3	multivariate
Elution flow rate	cm/hr	200	100-300	150-250	univariate

Each multivariate factor also carries a coded letter, which is given in Table 6.2 and used in place of the parameter name in the planned design matrices.

Table 6.2: Coded letters for the multivariate factors, used in Table 14.1 and Table 14.2.

Code	Factor	Unit	Studied in
A	Protein load	g/L resin	screening + RSM
B	Load/Wash conductivity	mS/cm	screening + RSM
C	Elution buffer pH	pH	screening + RSM
D	Elution stop collect	OD desc.	screening + RSM

The protein load will be studied from 10-40 g/L resin against a normal operating range of 18-32, and the load and wash conductivity from 3-7 mS/cm against a normal operating range of 4-6. Elution pH will be studied from 5.8-6.2 and the stop collect criterion from 0.5-1.5 on the descending edge of the elution peak. The set-point of every factor is the centre of its characterization range, so the centre point of both

designs is the nominal operating condition of the step, and the normal operating range of each factor lies inside the range this study will bound.

The knowledge space of the step is the box formed by the four multivariate characterization ranges. The elution flow rate is bounded on its own axis, over the range given in Section 6.4. The design space that PCR-007 defines will be a subset of that box, and the process will be operated inside a control space that lies inside the design space (CMC Biotech Working Group 2009). Nothing in this plan claims performance outside the box.

6.2 Screening design

The screening study is a two-level full factorial in the 4 multivariate factors with 3 centre point replicates, which is 19 runs in total. The coded matrix is given in Table 14.1, and the runs will be executed in randomized order (SOP-1002).

A full factorial was chosen in preference to a fractional design because 4 factors need only 16 factorial runs. Fractioning would save little and would alias interactions with main effects, and the mechanisms set out in Section 4.1 predict interactions between load and both of the parameters that control selectivity. All 4 main effects and all 6 two factor interactions are estimated free of each other (Table 14.1).

The purpose of this design is to identify the factors that matter, which it does with runs at the edges of the knowledge space. It cannot describe curvature, because a two-level design has no information between its levels. The centre point replicates supply that information in aggregate: a difference between the mean of the centre points and the mean of the factorial runs is evidence of curvature, and it is the first evidence that the quadratic terms of the response surface design are needed. The screening model is not used to define an operating region under any outcome.

6.3 Response-surface design

The response surface study is a face-centred central composite design in the same 4 factors. It comprises 16 factorial runs, 8 axial runs and 4 centre point replicates, which is 28 runs in total (Table 14.2). The axial points sit on the faces of the cube, so no run leaves the characterization range of any factor. That choice keeps every run inside the ranges the scale-down model and the platform buffer system support, at the cost of a design that is not rotatable.

This design estimates the full quadratic model described in Section 5.5. It is the predictive model of the step, and it is the basis of the design space and of the proven acceptable ranges reported in PCR-007.

All 4 screening factors are carried into the response surface design. They are carried because RA-001 scored each of them against a quality attribute (Table 4.2), and because a factor dropped after screening cannot be recovered later without a new study. Where the screening data show a factor to be inactive, that result is reported,

and the factor stays in the response surface model as evidence that the step is robust to it across the range studied.

6.4 Univariate assessment

Elution flow rate will be assessed one at a time, with every other parameter held at its set-point. It will be run at the edges and at the centre of its characterization range of 100–300 cm/hr, with replication at the set-point, and it will be assessed against the same three responses and the same acceptance criteria as the multivariate runs.

Flow rate was kept out of the multivariate design for a mechanistic reason. It acts through residence time and pressure drop, so its effect on the separation is expected to be monotonic and small inside the range studied, and it is not expected to interact with the parameters that set selectivity. A parameter assessed one at a time can support a proven acceptable range, and that is what these runs are for. The range is read from the measured responses of the runs themselves, by the rule given in Section 8, because the flow rate is not a factor of the fitted model. It cannot support a claim about interactions, and no such claim will be made for it (CMC Biotech Working Group 2009).

7 Acceptance and decision criteria

This section fixes the decision rules before the study is executed. Five decisions will be made in the course of the work, and each has its criterion here. A decision that cannot be made against the criterion stated here is escalated, documented and approved in the same way as an amendment to this plan (Section 13).

Scale-down model qualification. The model is qualified when triplicate runs at the set-point show no statistically significant difference from the commercial scale values of yield, aggregate clearance, HCP clearance and pool volume at $\alpha = 0.05$, and when column efficiency meets the packing criteria of SOP-2008. A comparison on triplicate runs has little power, so the absence of a significant difference is not on its own evidence of equivalence. The magnitude of each difference is therefore reported with the test, and the risk that a small difference escapes detection at this sample size is carried in Section 12. No designed experiment starts before the model is qualified, and a failure of qualification stops the study until it has been investigated and the model has been corrected.

Execution. A design is accepted for analysis when every run was executed at its planned settings, when the centre point replicates are consistent with the reproducibility expected of the step, and when every sample was taken and assayed as described in Section 5.4. Deviations will be recorded, investigated, impact assessed and reported in PCR-007 with their disposition. Where a run was executed at a setting other than the planned one, and the executed setting lies inside the characterization range, the analysis will use the executed value.

Model adequacy. A response surface model is adequate when its overall F test is significant at $\alpha = 0.05$, when the lack of fit test against pure error is not significant at $\alpha = 0.05$, and when its residuals show no structure against the predicted values. A model is used for prediction only when its predicted R^2 is in reasonable agreement with its adjusted R^2 , which is the test of whether it will predict a run it has not seen. Where a response fails these tests, PCR-007 will say so and that response will not be used to define the operating region.

Operating region. The operating region is acceptable when the fitted response surface model predicts pool aggregate at or below 5 % HMW (SEC) across the whole normal operating range of every parameter, and when the region defined in that way contains the set-point with margin on every axis. Step yield is a process performance attribute and not a quality attribute. It is reported for every run and is used to judge the consistency of the step, and it does not by itself constrain the operating region.

Host cell protein is treated differently, and the reason is stated plainly here because it decides how a large part of the study will be read. The drug substance criterion of 100 ng/mg applies to the drug substance and not to the pool of an intermediate step. Anion

exchange provides further HCP clearance downstream of this step (PCP-008), so a pool HCP level above the drug substance criterion is not by itself a failure of this step. This plan therefore sets no in-process limit on pool HCP. It commits instead to reporting the fitted model of pool HCP with its ranges, and to carrying that model into the anion exchange characterization and the roll-up in PCMR-001, where the clearance of the three steps is assessed together against the drug substance criterion.

Parameter classification. Each parameter will be classified under SOP-4001 from the effects demonstrated in this study and from the risk that it falls outside the design space in routine operation. A parameter with a demonstrated effect on a critical quality attribute is classified as critical, and as well controlled critical where the control capability of the commercial equipment makes an excursion unlikely. A parameter with no demonstrated effect is classified as a general process parameter, and the evidence of its lack of effect is retained in the knowledge space as evidence of robustness across the range studied (CMC Biotech Working Group 2009).

8 Proven acceptable ranges (planned analysis)

A proven acceptable range will be reported for each governed attribute and each multivariate parameter, and for the elution flow rate from its own runs. The proven acceptable range is a univariate statement about one parameter, and the design space is a multivariate region. Both are reported because they answer different questions, and because a range proven with the other parameters held still is not the same assurance as a region proven with them moving.

The acceptance basis is the drug substance criterion of the attribute, taken from the same register that produced Table 4.1, so that the criterion used in the calculation is the criterion the process is held to. No back calculation is needed at this step. A viral clearance response would need one, because the log reduction required of a single step is only defined once the clearance credited to the other steps has been subtracted from the cumulative requirement, but this step carries no viral clearance claim (Section 1).

The two governed responses are not read the same way against that basis. The criterion for aggregate binds at this step, because no later step reduces aggregate, so an aggregate range is a range the commercial process is held to. The criterion for HCP does not bind here, because Section 7 sets no step level limit on the pool of an intermediate step. The HCP ranges are computed against the drug substance criterion for information, and they feed the combined assessment of the three clearance steps in PCP-008 and PCMR-001 instead of a pass or fail at this step.

Two analyses will be run for each multivariate parameter and each attribute.

- The first holds the other parameters at their set-points and evaluates the fitted response surface model across the characterization range of the parameter of interest.
- The second propagates the normal operating ranges of the other parameters. It evaluates the model on a grid of 81 points across that same range, and at each grid point 2,000 batches are drawn with the other parameters distributed inside their normal operating ranges. The residual variability of the model is added to every draw, and a 95 % predictive interval of the attribute is formed at each grid point.

The decision rule is the same in both analyses. The proven acceptable range is the contiguous interval of the parameter, containing the set-point, over which the attribute meets its acceptance criterion. In the first analysis that criterion is applied to the predicted mean. In the second it is applied to the whole 95 % predictive interval,

which is a robustness criterion, so the second range is expected to be the narrower of the two.

The elution flow rate is not a factor of the fitted model, so neither analysis applies to it. Its range is read from the measured responses of the univariate runs described in Section 6.4, as the interval of the flow rate over which those responses meet the same acceptance basis, with every other parameter at its set-point. The range rests on those runs alone. It carries no statement about what happens when the other parameters move, and PCR-007 will report it on that footing.

Each range derived from the fitted model will be reported with a plot of two panels, one for each analysis. The parameter is on the horizontal axis and the attribute on the vertical axis, the acceptance limit is drawn, the acceptable region is shaded green, and the set-point and the normal operating range are marked.

Three outcomes are possible for each pair analysed against the fitted model, and PCR-007 will report which of them applies. Where the whole characterization range is acceptable under both analyses, the parameter is proven acceptable across the range studied and robust to variation of the others inside their normal operating ranges; the constraint that then binds the operating region is the multivariate design space corner and not any univariate range. Where the second analysis gives the narrower range, the report will name the attribute that binds it and state the margin that remains between the normal operating range and the edge of the proven range. Where no interval containing the set-point meets the criterion, the report will state that plainly, and will assess the attribute against the clearance credited to the downstream steps as set out in Section 7.

Every proven acceptable range reported from this study is bounded by the characterization range that produced it, and by the fitted model or the runs it was read from. None of them is an extrapolation, and none of them is confirmed at commercial scale by this study.

9 Data management and integrity

Data from this study are managed under the ALCOA+ principles. Chromatograms and instrument data are captured in the chromatography data system with the audit trail enabled, and analytical results are entered in the laboratory information management system against the sample identity assigned at the time of sampling. Executed run settings are recorded from the instrument record and not from the plan.

The statistical analysis is performed under SOP-1002 in a scripted environment, so that the whole analysis can be re-executed from the raw data without a manual step. Every table and figure in PCR-007 is produced by that analysis. Raw data, the analysis scripts and the fitted models are retained for the life of the product under the record retention procedure of the quality system.

A second person reviews the sample registration, the executed design settings and the analysis output before the results are used. Changes made after that review are documented under change control with the reason for the change.

10 Roles and responsibilities

The study is executed by Process Development with support from the functions listed in Table 10.1. The approval record in the front matter names the functions that approve this plan.

Table 10.1: Functions and their responsibilities for the study.

Function	Responsibility
Process Development / MSAT	Owens this plan, executes the runs, performs the statistical analysis and authors PCR-007
Analytical Development	Provides the validated methods, analyses the samples and assesses any method-related finding
Manufacturing Science	Confirms that the operation modelled is the operation at commercial scale and reviews the resulting ranges
Statistics	Reviews the designs before execution and the fitted models before they are used to define ranges
Quality Assurance	Approves this plan and PCR-007, and oversees the handling of deviations

11 Deliverables and schedule

The deliverable of this plan is PCR-007, which reports the executed study, the fitted models, the design space, the proven acceptable ranges and the classification of every parameter. PCR-007 rolls up into PCMR-001.

The work runs in the phases listed in Table 11.1, in that order. No calendar dates are fixed here. The sequence is fixed, because no design may be executed on an unqualified model, and because the response surface design is executed only once the screening data have confirmed that the ranges and the assumptions of the design still hold.

Table 11.1: Phases of the study, their outputs and their preconditions.

Activity	Output	Precondition
Scale-down model qualification	Qualification summary filed under SOP-1001	Column packed and qualified under SOP-2008
Screening design execution and analysis	Effect estimates for each response	Model qualified
Response surface execution and analysis	Fitted models, operating region, proven acceptable ranges	Screening analysed and reviewed
Univariate assessment of elution flow rate	Proven acceptable range for the univariate parameter	Model qualified
Reporting	PCR-007	All phases complete and reviewed

12 Risks and assumptions

Four risks could affect the conclusions of this study, and each has a control.

Scale-down representativeness. The study assumes that a qualified small column behaves like the commercial column over the whole characterization range, while the qualification evidence is generated at the set-point. The control is the qualification described in Section 5.1 and its acceptance criterion in Section 7. The residual risk is that a difference too small to detect in triplicate qualification runs grows at an edge of the knowledge space. Design space edges are not confirmed at commercial scale by this study.

Feed representativeness. The load is an intermediate of the same process, so its aggregate and HCP content depend on how the upstream steps were run. A feed that is not representative displaces the intercept of every fitted model, and the model then describes material the commercial process will not produce. The control is characterization of each feed lot before use, and the use of one lot for all runs of a design wherever the material allows it. PCR-007 will state which lots were used.

Assay variability. About 40 % of the variance of a pool HCP result comes from the method (Table 5.1). The control is the replicated centre point, which separates that variance from lack of fit, and the interpretation rule in Section 5.3, which attributes near-limit differences in HCP to the assay unless the design shows otherwise.

Material and schedule. Resin, buffers and feed have to be available for the runs of a design to be executed close together. The control is that runs within a design are randomized and executed in one campaign wherever possible, so that a break in the campaign is not confounded with a factor setting.

The plan makes three assumptions and states them here so that they can be challenged. The step does not modify the charge variant or glycan attributes, which are not measured in this study. The resin is within its validated cycle life, and cycle number is held constant and is not a factor of the design. The step makes no viral clearance claim, so no small scale spiking study is run here.

13 Approvals

This plan takes effect when it has been approved by the functions listed in the approval record in the front matter, and that record is the authority for executing the study. An amendment to a design, to a range or to an acceptance criterion after approval is documented, justified against the risk basis in Section [4.3](#) and approved in the same way, before the affected runs are executed.

14 Appendices

The planned design matrices are given below. No table in this plan carries a result, because no run has been executed at the time of approval.

14.1 Appendix A — Planned screening design matrix

The coded settings of the 19 screening runs are given in Table 14.1. Coded -1 and +1 are the edges of the characterization range of a factor and 0 is its set-point, and the letters are those of Table 6.2.

Table 14.1: Planned screening design matrix, coded settings.

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	center	0	0	0	0
18	center	0	0	0	0
19	center	0	0	0	0

The natural setting of each run follows from Table 6.1 and Table 6.2, and the run sheets issued for execution carry it in natural units.

14.2 Appendix B — Planned response-surface design matrix

The coded settings of the 28 response surface runs are given in Table 14.2. The axial runs are the rows with one factor at an edge and the others at 0.

Table 14.2: Planned response-surface design matrix, coded settings.

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	-1	0
22	axial	0	0	1	0
23	axial	0	0	0	-1
24	axial	0	0	0	1
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0

References

- CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.
- International Council for Harmonisation. 2009. *ICH Q8(R2): Pharmaceutical Development*. ICH.
- International Council for Harmonisation. 2012. *ICH Q11: Development and Manufacture of Drug Substances*. ICH.
- International Council for Harmonisation. 2023a. *ICH Q5A(R2): Viral Safety Evaluation of Biotechnology Products Derived from Cell Lines of Human or Animal Origin*. ICH.
- International Council for Harmonisation. 2023b. *ICH Q9(R1): Quality Risk Management*. ICH.
- U.S. Food and Drug Administration. 2011. *Guidance for Industry: Process Validation — General Principles and Practices*. FDA.